

Chapter 8: Determinants Worksheet

1. Are each of the following statements true or false?

- (a) If A, B are 2×2 matrices, then $\det(A + B) = \det(A) + \det(B)$.
- (b) If M is a 2×2 matrix with $\det(M) = 0$, then at least one of the entries of M must be zero.
- (c) $\det(2I_3) = 2$.

2. Compute the following determinants using cofactor expansion. (Recall that $\det(A) = |A|$).

(a) $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$

(b) $\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix}$

(c) $\begin{vmatrix} -3 & 2 \\ 4 & 1 \end{vmatrix}$

(d) $\begin{vmatrix} 5 & 3 \\ 10 & 6 \end{vmatrix}$

(e) $\begin{vmatrix} 4 & 1 & 3 \\ 2 & 3 & 0 \\ 1 & 3 & 2 \end{vmatrix}$

(f) $\begin{vmatrix} 2 & 3 & 1 \\ 3 & 5 & 2 \\ 1 & 2 & 1 \end{vmatrix}$

(g) $\begin{vmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 3 & 2 & 0 \\ 4 & 0 & 2 & 1 \end{vmatrix}$

3.

(a) Find all values of k in $A = \begin{bmatrix} k & 5 \\ 4 & k+1 \end{bmatrix}$ such that $\det(A) = 0$.

(b) Find all values of x in $M = \begin{bmatrix} x & 0 & 0 \\ 1 & x-2 & -2 \\ 4 & 2 & x+3 \end{bmatrix}$ such that $\det(M) = 0$.